Assessing the Correlation Between Social Media Usage and Mental Well-Being

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Abstract— ***The study examined the relationship between social media usage and the mental well-being of university students. A quantitative cross-sectional design was used to collect data from 100 college students' peers selected through convenience sampling. Data were gathered through a questionnaire survey consisting of two parts: the first part measured average daily social media usage in hours, and the second assessed mental well-being scores using a validated 5-item Likert Scale. Correlation analysis revealed a strong negative relationship between social media usage and mental well-being, implying that excessive screen time is associated with lower psychological well-being among students. These findings highlight the critical need for digital wellness programs within institutions and provide a foundation for developing mental health support systems in higher education.***

Keywords— ***Social media usage, mental well-being, cross-sectional design, digital wellness, higher education***

I. Introduction

Social media has been part of Filipinos' everyday lives, influencing how we interact, communicate, function, and spend our time. Students widely use platforms like Facebook, TikTok, and Instagram for various purposes, including education, communication, and entertainment. According to an article, nearly 81.9% of the population of the Philippines is active, with approximately 95.8 million social media users [1]. However, social media can affect your mental well-being and happiness; excessive use can change how you feel, cause dissatisfaction, isolation, and build up depression, anxiety, and stress [2].

Social media like Instagram and Facebook are among the social media that most of the Filipino youth and adults are now using for a long period of time. [3]. However, concern regarding using social media is how this app impacts the mental well-being of the Filipino youth. Meanwhile, according to an article, numerous studies on social media's effects have been conducted, and it has been proposed that long-term use of social media like Facebook may be linked to negative manifestations and symptoms of depression, anxiety, and stress. The problem, however, is that the risk factors of being bullied and social isolation are readily available at this period [4].

The growth of social media affected numerous people. In the twenty-first century, Filipino youth use social media as their most apparent characteristic. Whether at home, school, or work, the Filipino youth are constantly connected through Social media accounts. According to an article, a 2025 review of research found that one of the most used social media platforms, which is Instagram, has a user who develops depression, anxiety, and decreased self-esteem [5].

In this regard, the goals of the researchers are to study the relationship between social media usage and the mental well-being of the students at different univeristy. Using a cross-sectional research design and a questionnaire survey, the researchers aim to examine the correlation between the two variables. It is hoped that the results will be beneficial for the university students and future researchers in understanding how social media can affect the mental well-being of every Filipino youth.

II. Methodology

This study was conducted in order to assess the connection between social media usage and a person's mental well-being. As such reasons stated, the study employs a correlational research design. Through this medium, it allows the researchers to characterize the current state of both variables and to determine whether a significant relationship exists between the two entities.

The respondents of this study consisted of 100 college student peers, ranging from first to fourth-year students. The usage of convenience sampling was imposed in selecting participants for this study. Through this, any student who is available and possesses an amount of social media usage is considered eligible to participate in the study.

Data from this study were collected through the use of an online survey platform hosted by Microsoft. It is commonly referred to as Office Forms; the form uses a tailored questionnaire consisting of two segments. The first portion of the survey measures the respondent's frequent usage of social media platforms. The second portion of the survey administers a five-point Likert scale to measure the influence of social media in relation to the respondent's mental well-being. Items within the Likert scale scales within the range from 1(Strongly Disagree) to 5(Strongly Agree).

The gathered data were analyzed using the components of descriptive statistics and inferential statistics, imposing the use of Mean, Graphs, Pearson's r and Linear Regression. These instruments are responsible for addressing the nature of social media usage to a person's wellbeing. The utilization of Pearson's Product-Moment Correlation Coefficient, also known as "Pearson's r," is responsible for determining the strength and direction of the relationship between two continuous variables. Additionally, charts are used to visually demonstrate the distribution of the collected responses. The results were summarized using descriptive statistics (mean, standard deviation) to figure out the levels of hours spent on social media platforms. To impose the relationship between social media usage and mental wellbeing, Pearson's r is applied to each part of the survey that employs the Likert scale. Furthermore, linear regression analysis was used to model the relation between the variables; this analysis takes the form of a scatterplot. All data for this study were processed using Python libraries, such as pandas, matplotlib, Counter, & linregress. The significance level was set at $p < 0.05$.

III. Results and discussion

To assess the relationship between social media usage hours and students' mental well-being, descriptive and statistical analyses were carried out. The datasets included mental well-being scores, time spent using social media, and demographic data gathered from the respondents. The analysis demonstrated that students' excessive screen time is significantly associated with lower mental well-being scores. A series of visualizations was built to help people better understand the correlation between social media usage and psychological well-being. These charts illustrate important patterns and relationships that validate the study's results and provide insight into the impact of social media on students' mental well-being. A comprehensive description of each visualization follows.

Table I. presents the demographic profile of the respondents in terms of gender and age group. The result shows that the majority of the participants were male students, comprising 63% of the total population, while only 37% were female. In terms of age distribution, most participants belonged to the 20-21 group, representing 64% of the respondents. Meanwhile, 30% were aged 18-19 and only 6% belonged on the group of 22-23.

Variables	*Category*	*Frequency*	*Percentage*
Gender	Male	63	63%
	Female	37	37%
Age	18-19	30	30%
	20-21	64	64%
	22-23	6	6%

Table 1 Demographic Profile

Figure I. This pie chart displays the gender distribution of the participants. The results show that 63% of the participants were male, while 37% were female. This indicates that the majority of the respondents were male students.

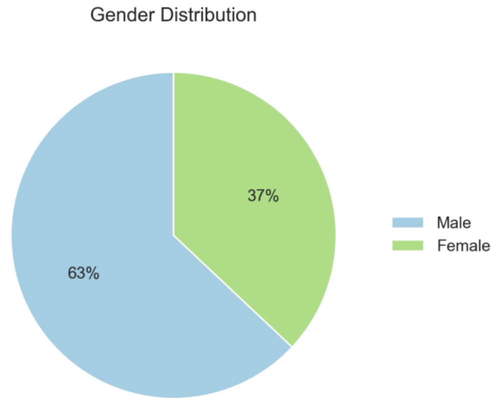

Figure1. Gender Distribution of Participants

Figure II. The histogram shows how many hours students spend on social media each day. Most of the students cluster around 5 to 6 hours daily. While only a few students keep their usage low at 4 hours and below, one notable group uses social media for 8 hours and above. This proves that a daily part of a student's life is focused on screen time.

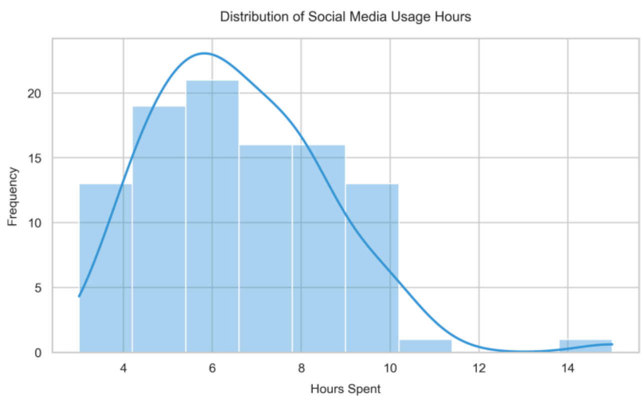

Figure2. Distribution of Social Media Usage Hours

Figure III. The well-being scores chart shows 2 unique peaks in shape rather than a normal curve. Student scores are split into two groups: one group clustering around 13.5 and another group clustering around a higher score of 18.5. This shows a clear divide between students who are doing well emotionally and those who are struggling.

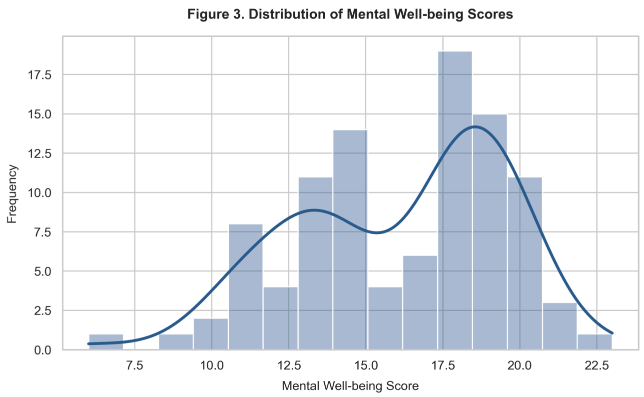

Figure3. Distribution of Mental Well-Being Score

Table II. presents the descriptive statistics of the study variables. The respondents spent an average of 6.61 hours per day on social media (SD = 1.97). Meanwhile, the mean mental well-being score was 16.13 (SD = 3.40). These results indicate moderate social media usage and mental well-being levels among the respondents.

Variable	*Mean*	*SD*
Social Media Usage Hours	6.61	1.97
Mental Well-being Score	16.13	3.40

Table 2 Descriptive Statistics

Table III. presents the Pearson correlation analysis between social media usage hours and mental well-being scores. The computed r-value of -0.91 indicates a strong negative relationship between the two variables. This suggests that as social media usage hours increase, students' mental well-being scores tend to decrease.

Variables	*r-value*	*Interpretation*
Hours vs Well-being	-0.91	Strong negative

Table 3 Pearson Correlation Analysis

Figure IV. This scatter plot displays the correlation between average hours spent on social media (x-axis) and mental well-being score (y-axis) among 100 participants. The data reveals a strong negative correlation (*r* = -0.91)

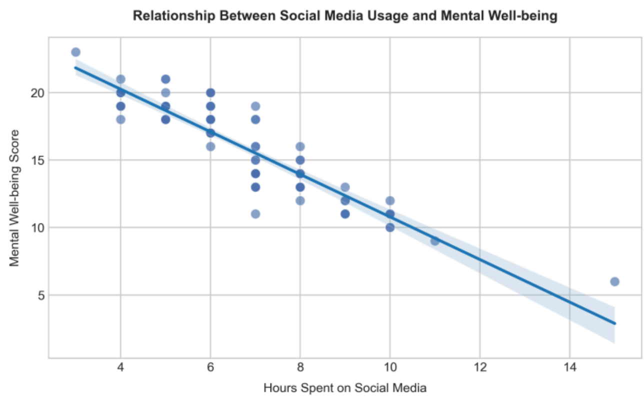

Figure4. Relationship Between Social Media Usage and Mental Well-Being

Figure V. The Correlational Matrix shows a massive negative correlation of -0.91 between daily hours and total well-being scores. A number this close to -1.0 means the two variables are powerfully linked. This shows that daily hours negatively impact all five individual survey questions. This proves that excessive social media usage damages multiple parts of a student's life, including focus, mood, and sleep

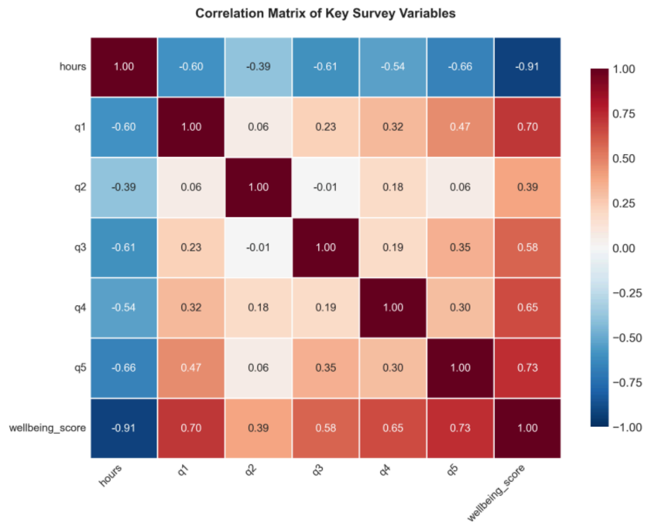

Figure5. Correlation Matrix of Key Survey Variables

Figure VI. The box plot compares male and female well-being scores. The median score is exactly the same (17) for both genders, meaning social media affects the average male and female student similarly. However, the male group shows a much wider spread of scores. Meaning a specific group of male students is highly vulnerable to severe mental health drops.

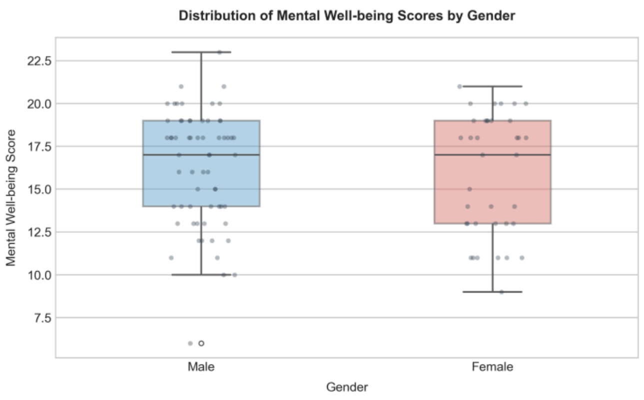

Figure6. Mental Well-Being Scores by Gender

Figure VII. This line graph illustrates the average mental well-being score corresponding to different levels of daily social media usage. The results show that students who spend fewer hours on social media generally report higher mental well-being scores. As the number of hours spent on social media increases, the average mental well-being score gradually declines. This trend suggests a negative relationship between social media usage and mental well-being.

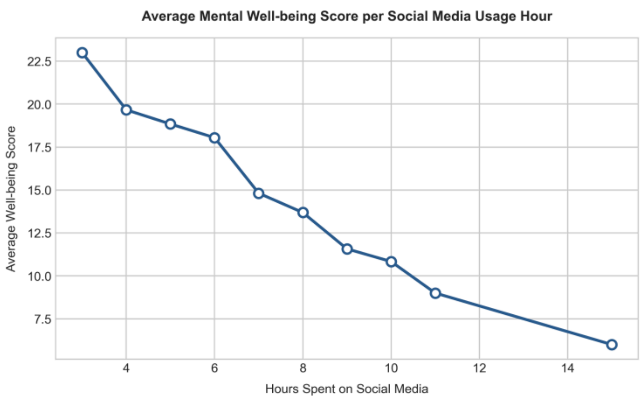

Figure7. Average Mental Well-Being Score by Social Media Usage Hours

Table IV. presents the results of the linear regression analysis between social media usage hours and mental well-being scores. The regression equation, $y = 26.57 - 1.58x$, indicates that for every additional hour spent on social media, the mental well-being score decreases by approximately 1.58 points. The negative slope and the strong negative correlation ($r = -0.91$) suggest an inverse relationship between the two variables. Furthermore, the p-value (< 0.001) indicates that the relationship is statistically significant.

Parameter	***Values***
Slope (b)	-1.58
Intercept (a)	26.57
Pearson's r	-0.91
p-value	<0.001
Regression Equation	**y = 26.57 - 1.58x**

Table 4 Linear Regression Analysis

IV. Conclusion

This research paper or study aimed to determine the effects and relationship or correlation of social media use and the mental health well-being of university students. In accordance with the gathered survey and questionnaire, the researchers were able to analyze the respondents' answers and identify important findings related to the analysis paper. The result of the gender distribution of participants is summed up by 63% of the participants being male, while the other 37% are female. For the Distribution of Social Media Usage Hours, the data analysis results show the mean is 6.61. In the Mental Well-being Score, the mean yields a value of 16.13. Further analysis of the correlation between hours of use and well-being indicates an inverse relationship; this is due to the r-value being -0.91, which implies a strong negative. With regards to gender being a factor for mental well-being, based on the mean of both genders having a small difference, the information suggests that gender has little to no impact on an individual's mental well-being. The study suggests that lowering time with social media use may benefit the mental well-being of an individual, regardless of gender, since the study has concluded that the correlation between the hours of use of social media and mental well-being is inversely related.

References

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APPENDICES

Appendix A. Survey questionnaire

Part I. Demographic Information

1. Gender
- Male
- Female
2. Age: ______
3. What social media platforms do you use most?
4. On average, how many hours per day do you spend on social media?

Part II. Mental Well-being Assessment

Rating Scale:

Scale	Description
1	Strongly Disagree
2	Disagree
3	Neutral
4	Agree
5	Strongly Agree

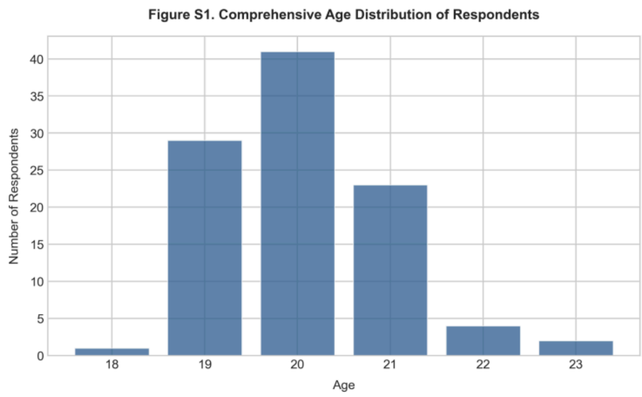
Statements:

1. I feel good about myself after using social media.
2. I can focus on my daily tasks without worrying about missing out on social media activities (FOMO).
3. My sleep quality is not negatively affected by my social media use at night.
4. I rarely feel stressed or overwhelmed by the content I see on social media.
5. I remain confident and satisfied with my life, even when comparing myself to others online

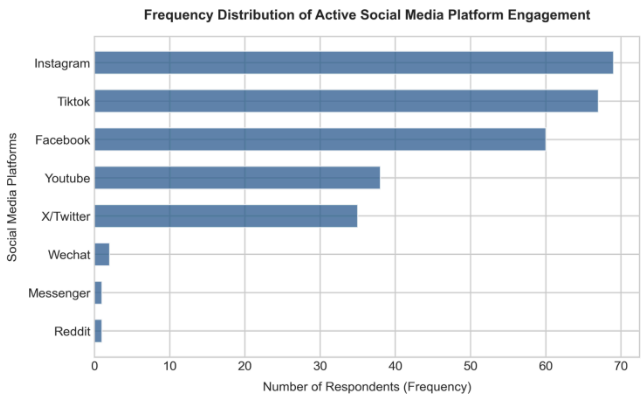
Appendix B. Raw Dataset

1	Gender	Age	What social	On average	Mental Well-being	Mental Well-being	Mental Well-being	Mental Well-being	Mental Well-being
2	Male	21	Youtube;	7	Strongly Di	Neutral	Strongly Ag	Agree	Disagree
3	Male	20	Instagram;T	5	Agree	Neutral	Agree	Neutral	Strongly Agree
4	Female	20	Tiktok;Face	5	Neutral	Neutral	Strongly Ag	Agree	Agree
5	Male	19	Youtube	5	Neutral	Neutral	Strongly Ag	Agree	Neutral
6	Male	20	Facebook;Y	4	Agree	Agree	Agree	Strongly Ag	Neutral
7	Male	20	Facebook;T	5	Neutral	Agree	Agree	Neutral	Agree
8	Male	20	Instagram;T	8	Disagree	Neutral	Disagree	Neutral	Agree
9	Female	19	Tiktok;Face	6	Neutral	Agree	Neutral	Agree	Agree
10	Male	20	Tiktok;Insta	7	Disagree	Strongly Ag	Disagree	Strongly Ag	Disagree
11	Male	20	Facebook;T	10	Disagree	Neutral	Neutral	Strongly Di	Strongly Disagree
12	Male	20	Instagram;Y	6	Agree	Neutral	Agree	Agree	Neutral
13	Female	19	Facebook;T	9	Disagree	Agree	Strongly Di	Disagree	Disagree
14	Male	19	Instagram	6	Agree	Neutral	Agree	Neutral	Agree
15	Female	20	Tiktok;Insta	9	Disagree	Agree	Strongly Di	Disagree	Disagree
16	Male	20	Facebook;T	4	Agree	Agree	Agree	Agree	Agree
17	Female	20	Facebook;In	8	Neutral	Agree	Neutral	Neutral	Strongly Disagree
18	Male	21	Instagram	4	Strongly Ag	Neutral	Agree	Agree	Agree
19	Female	18	Tiktok;Insta	8	Agree	Disagree	Disagree	Agree	Disagree
20	Female	21	Youtube;In	9	Strongly Di	Neutral	Disagree	Disagree	Neutral

Appendix C. Additional Charts



FigureC1. Age Distribution of Respondents



FigureC2. Distribution of Used Social Media Platforms

Appendix D. Statistical Computations

Pearson's r Formula

$$r = \frac{\Sigma((X_i - \bar{X}) * (Y_i - \bar{Y}))}{\sqrt{\Sigma(X_i - \bar{X})^2 * \Sigma(Y_i - \bar{Y})^2}}$$

where:
r = Pearson correlation coefficient
n = number of respondents
Σxy = sum of the product of x and y
Σx = sum of x-values
Σy = sum of y-values
Σx^2 = sum of squared x-values
Σy^2 = sum of squared y-values

Linear Regression Formula

y = a + bx
where:
y = predicted mental well-being score
a = intercept
b = slope
x = social media usage hours